Team Members

- Bhavy Patel
- Harsh Shah
- Pooja Kondath
- Rishi Shah

Dataset Selection

IPL Complete Dataset (2008-2024)
This dataset provides ball-by-ball and match-level data from the Indian Premier League (IPL) between 2008 and 2024. It includes match outcomes, venues, teams, toss results, player awards (matches.csv), and detailed delivery information like batters, bowlers, runs, extras, and dismissals (deliveries.csv). It's ideal for analyzing team performance, player stats, and match trends.

Spotify - All Time Top 2000s Mega Dataset
This dataset contains detailed audio and popularity information about songs from various genres and artists. Each row represents a song and includes its title, artist, genre, release year, and audio features such as energy, danceability, loudness, and popularity. It's useful for analyzing trends in music characteristics across time and genres.

WDI: Education, Health & Employment (2011-2021)
This dataset, compiled from the World Bank's World Development Indicators (WDI), covers 19 advanced economiesbetween 2011 and 2021. It includes over 80+ socioeconomic indicators on education, healthcare, employment, research, and population structure — supporting international comparisons, policymaking, research, and forecasting.

Understanding Selected Dataset

IPL Complete Dataset (2008-2024)

matches.csv

- **match_type** — Categorical: Type of Match (e.g. League, Semi Final)
- **city** — Categorical: City where match was played
- **venue** — Categorical: Stadium/Venue name
- **team1** — Categorical: First Team
- **team2** — Categorical: Second Team
- **toss_winner** — Categorical: Toss-winning team
- **toss_decision** — Categorical: Toss decision (Bat or Field)
- **winner** — Categorical: Match-winning team
- **result** — Categorical: Type of result (runs/wickets/tie)
- **super_over** — Categorical: Whether super over was used(Y/N)
- **method** — Categorical: Method used to declare result (e.g. D/L method)
- **player_of_match** — Categorical: Player awarded for the best performance
- **umpire1** — Categorical: First Umpire
- **umpire2** — Categorical: Second Umpire
- **id** — Ordinal: Unique match ID
- **season** — Ordinal: Season Year (e.g 2008,2009)
- **target_overs** — Ordinal: Overs allotted in second innings
- **target_runs** — Ratio: Target score for second innings
- **result_margin** — Ratio: Margin of score
- **date** — Interval: Match Date

matches.csv

- **batting_team** — Categorical: Team batting during the delivery
- **bowling_team** — Categorical: Team bowling during the delivery
- **batter** — Categorical: Name of the batter facing the delivery
- **bowler** — Categorical: Name of the bowler delivering the ball
- **non_striker** — Categorical: Non-striking batter
- **player_dismissed** — Categorical: Name of player dismissed
- **dismissal_kind** — Categorical: Type of dismissal (e.g., bowled, caught, run out)
- **fielder** — Categorical: Fielder involved in dismissal
- **is_wicket** — Categorical: 1 if it resulted in a wicket, 0 otherwise
- **extra_type** — Categorical: Type of extra (legbyes, wides, no ball, etc.)
- **match_id** — Ordinal: Unique match ID
- **inning** — Ordinal: Inning number (1 or 2 or 3)
- **over** — Ordinal: Over number within the inning
- **ball** — Ordinal: ball number within the over
- **batsman_runs** — Ratio: Runs scored by the batter
- **extra_runs** — Ratio: Extra runs (wide, no balls, etc.)
- **total_runs** — Ratio: Total runs from the delivery(batter+extras)

Potential Dashboard Users

- **Cricket Analyst & Commentators** — Cricket analysts and commentators rely on historical data to extract insights about team performance, player statistics, and match dynamics. The dashboard would help them deliver informed commentary and conduct deep match analysis before, during, or after games.
- **IPL Teams & Coaches** — Teams and coaching staff use performance analytics to shape match strategies, assess opposition tactics, and monitor player development. The dashboard enables informed tactical planning and supports player selection, training focus, and in-game decision-making.
- **Fantasy Cricket Players & Apps** — Fantasy sports users and developers need real performance metrics to create competitive fantasy teams and recommendation engines. The dashboard helps track player form, consistency, and situational performance for better decision-making
- **Students, Educators & Researchers** — In academic settings, students and educators use sports data to learn concepts in statistics, data visualization, and machine learning. The dashboard provides an engaging, real-world dataset for classroom exercises and projects.
- **Sports Journalists & Bloggers** — Journalists and bloggers use sports data to discover emerging trends, records, and stories worth covering. The dashboard allows them to explore season-wide comparisons, team records, and player form to produce engaging and insightful content

Potential Questions

- Which player has won the most "Player of the Match" awards across all seasons?
- What is the average win margin by runs and wickets for each team?
- How has each team performed when batting first vs chasing?
- What is the team's win percentage when winning the toss and choosing to bat or field?
- How do players perform in powerplay vs death overs (e.g., economy rate, strike rate)?
- Which bowlers have taken the most wickets in critical overs (16–20)?
- Which batters consistently score 30+ runs per match?
- Who are the most economical bowlers in the last three seasons?
- Which players perform best in Super Over matches?
- What are the correlations between toss decision, batting order, and winning the match?
- Are certain cities or venues more favorable to a specific team?
- Which city has hosted the most high-scoring matches?
- What trends exist in match results across seasons (e.g., more chases won over time)?
- Which teams have the highest head-to-head win ratio?

Preprocessing

id (from matches.csv) & match_id (from deliveries.csv)	change the name of "match_id" column to id from deliveries.csv to "id" and Perform an inner join to merge match-level and delivery-level data
over	Categorize into Powerplay (1–6), Middle Overs (7–15), and Death Overs (16–20)
player_dismissed, is_wicket	Replace "NA" with None; cast is_wicket to int for aggregation
toss_winner, toss_decision, team1, team2, winner	Derive whether a team batted first or chased using toss outcome logic
super_over	Filter where super_over = 'Y' for Super Over analysis
season	Filter records for only the last 3 seasons (2022–2024) as needed
city, match_type, venue, target_runs, target_overs, method, umpire1, umpire2	Remove these unused columns from matches.csv file
batsman_runs, extra_runs, total_runs	Aggregate runs per batter/match/season; calculate match totals and economy
bowler	Group by bowler for economy, wickets, and over-based performance
batter	Group by batter to compute total runs, average score, 30+ score counts
player_of_match	Group and count per player to find top awardees
result, result_margin	Normalize result types; convert result_margin to numeric for statistical analysis
toss_decision, winner	Calculate win rate correlation with toss decisions
inning, non_striker, extras_type, dismissal_kind, fielder	Remove these unused columns from deliveries.csv file

Dashboard Components

Which player has won the most "Player of the Match" awards?	Plot: Bar Chart (Top 10 players	• Pre-attentive Attributes: Length (bar height), Size • Plot Order: First (Top section – Player Achievements)
What is the average win margin (runs and wickets) for each team?	Plot: Grouped Bar Chart	• Pre-attentive Attributes: Length, Color • Plot Order: Second (Team Performance section)
How does each team perform while batting first versus chasing?	Plot: Plot: Stacked Bar Chart	• Pre-attentive Attributes: Length, Color • Plot Order: Third (Team Performance section)
What is the win percentage when choosing to bat or field?	Plot: Pie Chart	• Pre-attentive Attributes: Area (slice size), Color • Plot Order: Fourth (Match Strategy section)
How do players perform in powerplay vs death overs (strike rate, economy)?	Plot: Line Chart	• Pre-attentive Attributes: Position, Slope, Color • Plot Order: Fifth (Player Phase Performance section)
Which bowlers have taken the most wickets in overs 16–20?	Plot: Bar Chart	• Pre-attentive Attributes: Size, Length • Plot Order: Sixth (Bowling Performance section)
Which batters consistently score 30+ runs per match?	Plot: Bar Chart	• Pre-attentive Attributes: Size, Length • Plot Order: Seventh (Player Consistency section)
Who are the most economical bowlers in the last three seasons?	Plot: Scatter Plot	• Pre-attentive Attributes: Position, Color • Plot Order: Eighth (Bowling Efficiency section)
Which players perform best in Super Over matches?	Plot: Bar Chart	• Pre-attentive Attributes: Position, Length, Size, Color • Plot Order: Ninth (Special Situations – Super Overs)
Is there a correlation between toss decision and match outcome?	Plot: Stacked Bar Chart	• Pre-attentive Attributes: Position, Length, Size, Color • Plot Order: Tenth (Match Strategy section)